

Mohamed Ghaith Romdhani

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Summary

Final-year Bachelor's student in Networks and Systems Engineering with a strong interest in datacenters, enterprise networks, and IT infrastructure. Completed multiple internships within bank headquarters and datacenter environments. Experienced in networking, systems administration, cybersecurity (SIEM), AI, IoT, and sustainable technologies. Former IEEE Student Branch Chair, current mentor, and national challenge winner with proven leadership and team management skills

Education

Bachelor's student in Networks and Systems Engineering

- Higher Institute of Computer Science Mahdia(2023-2026)
- Focus: Networks, Systems, Datacenters, and IT Infrastructure
- End-of-Studies Project conducted with a bank

Academic & Professional Training

- Completed academic and professional development programs with **AMIDEAST**.
- Focus on English proficiency, communication, and international collaboration.

Experience

Infrastructure & Security Analysis AI Assistant — End-of-Studies Project

- Developed an offline AI assistant in collaboration with **AMEN Bank** for infrastructure monitoring and security log analysis.
- Integrated **Graylog** and **Zabbix** to collect and analyze logs, alerts, and monitoring data.
- Designed a local **RAG** system using internal infrastructure documentation and machine inventory data.
- Enabled natural language interaction for interpreting logs, investigating incidents, and assisting infrastructure teams.
- Built the solution with a fully offline architecture to meet enterprise security and confidentiality requirements.

IT & Datacenter Intern (SIEM), Bank Headquarters

- Worked on **Security Information and Event Management (SIEM)** using **Graylog**.
- Log ingestion, analysis, dashboards, and alerting for enterprise environments.
- Exposure to datacenter monitoring and security operations.

Founder, Clothing Brand

- Founded and managed an independent clothing brand.
- Managed branding, marketing, logistics, and customer relations.

Awards & Achievements

Winner — TN2056 National AgriTech Challenge

- Developed a smart hydroponic system designed to minimize water consumption and reduce human intervention.
- Built an MVP and presented the project at a national congress before an international jury.

Second Place — ICAIGE National Innovation Challenge

- Developed an innovative decorative plant prototype capable of converting solar energy into usable electrical power.

Third Place — INNOHACK National Challenge

- Developed an AI-assisted overtaking support system for buses and semi-trucks.
- Combined IoT, sensors, and vehicle-to-vehicle communication to generate a 3D environmental view and assist drivers in overtaking decisions using AI-based analysis.

Volunteering

IEEE Student Branch Chair, IEEE ISIMA SB

- Served as Chair for **1.5 years**, leading a multidisciplinary team.
- Led organization of workshops, conferences, competitions, and technical events.
- Won multiple **national-level IEEE challenges**.

IEEE Mentor, IEEE ISIMA Student Branch

- Mentoring students in technical projects, leadership, and IEEE activities.

Co-Lead, IEEEExtreme Tunisia

- Co-leading national IEEEExtreme initiatives and team preparation.

Active Member, Leo Club Carthage

- Participated in social, humanitarian, and community service initiatives.

Projects

LogI — Intelligent Infrastructure and Security Analysis Assistant

- Developed an AI-powered assistant for security log analysis in banking environments.
- Integrated Graylog, LLMs, and RAG techniques to analyze logs and generate security insights.
- Designed secure query generation and evidence-based incident analysis workflows.

SmartShield — SIEM Platform

- Designed and implemented a SIEM solution using Wazuh, TheHive, and Shuffle.
- Centralized log collection, event correlation, alert management, and automated incident response workflows.

Graylog SIEM Infrastructure

- Built and configured a centralized monitoring and log management platform using Graylog.
- Integrated multiple data sources for infrastructure monitoring and cybersecurity analysis.

AI Scope — AI-Powered Land Cover Dashboard

- Developed an AI-powered dashboard to track historical land cover changes from 2000 to 2023.
- Implemented machine learning models to forecast land cover dynamics up to 2033.
- Created interactive maps, trend visualizations, and an AI chatbot for insights and recommendations.

SmartAero — Smart Aeroponic System

- Developed an IoT-based smart aeroponic farming system for sustainable agriculture.
- Automated environmental monitoring and nutrient delivery processes.

EcoHarvest — Smart Solar Energy Solution

- Developed an innovative decorative plant prototype capable of converting solar energy into usable electrical power.
- Focused on combining sustainability, renewable energy, and smart design concepts.

Optix — AI-Assisted Overtaking System

- Developed an AI-assisted driving support system for safe overtaking of buses and semi-trucks.
- Combined IoT sensors, vehicle-to-vehicle communication, and AI analysis to generate a 3D environmental view for drivers.

FreeFire — Organic Waste to Gas System

- Designed a smart system that converts organic waste into usable gas for households and farms.
- Promoted sustainable energy production and waste reduction practices.

Through Their Eyes — Speech and Sign Language Translation System

- Developed a communication system capable of translating speech into sign language and vice versa.
- Designed to support communication and education for deaf and mute children.

Zima Green — Recycling Solution

- Built a plastic recycling solution capable of producing 3D printing filament from recycled materials.
- Manufactured customized 3D-printed keychains using recycled plastic waste.

Personal Portfolio Website

- Developed a personal portfolio website showcasing projects, technical skills, certifications, and achievements.
- Designed a responsive and modern user interface to highlight professional experience and technical work.

Technologies

Technical Skills

Cybersecurity & SIEM: Graylog, Wazuh, TheHive, Shuffle, SIEM Architecture, Threat Detection, Log Analysis, Incident Response, Security Monitoring, Event Correlation

AI, Data & Machine Learning: LLMs, LangChain, RAG, NLP, Whisper, Machine Learning, AI Chatbots, Data Visualization, Computer Vision, Speech Recognition, Recommendation Systems

Systems & Infrastructure: Linux, System Administration, Docker, Proxmox, VMware, VirtualBox, Zabbix, Cloud Concepts, Infrastructure Monitoring

Networking & IoT: Networking, Enterprise Networks, IoT Systems, Sensors, 3D Imaging sensors, Smart Systems, Embedded Systems

Development & Tools: Python, MongoDB, GitHub, REST APIs

Languages

Arabic (Native), English (Fluent), French (Fluent), Turkish (Conversational)